REPORT

Thematic Workshop 1

ICEG Person - Core Person Identification

Monday 21 September 2026 | 10:00 - 12:30

Marie-Elisabeth Belpaire (03.N.08) & Microsoft Teams (hybrid)

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PRACTICAL INFO

Date: 21/09/2026 (10:00 - 12:30)
Location: Marie-Elisabeth Belpaire building and via Microsoft Teams
Facilitator: Valy Van den Broeck (valy.vandenbroeck@vlaanderen.be)
Github Repository: https://github.com/belgif/thematic/tree/master/person

ATTENDEES

Name	Organisation
Jitse De Cock	Digitaal Vlaanderen
Amandine Pirnay	National Register (IBZ)
Bart Vrancken	National Register (IBZ)
Leatitia Roesems	Agentschap Landbouw en Zeevisserij
Suzanne Put	Digitaal Vlaanderen
Sofie Declerck	Federale Overheidsdienst Economie
Ellen Brems	Federale Overheidsdienst Economie
Filip Borloo	
Isabel Sastre Cantano	SMALS
Gert De Tant	Sirus
Nathan Claeys	KSZ - BCSS
Vincent Turine	KSZ - BCSS
Miguel Discart	KBO - BCE
Dylan Van Assche	Digitaal Vlaanderen
Lieven Verreycken	Digitaal Vlaanderen
Benoit Fricheteau	Paradigm Brussels
Sebastiaan Taes	BOSA

AGENDA

10:00 - 10:05	Welcome and practical information
10:05 - 10:15	Planning
10:15 - 10:20	Rationale for the trajectory and context
10:20 - 10:30	Approach and use cases
10:30 - 11:00	Storyline: Registration of a person
11:00 - 11:15	Break
11:15 - 11:30	Storyline: Registration of a noble title and pseudonym
11:30 - 12:15	Storyline: Change of name and registered sex
12:15 - 12:30	Q&A and next steps

INTRODUCTION

CONTEXT

Interoperability is the ability of different autonomous organisations or systems to communicate and work together. In the Belgian context this means that federal organisations, the Regions and the Communities must be able to share person-related data in such a way that all systems can work together and interpret the data in the same manner. Today, person data is not yet exchanged in a uniform way. Several problems were highlighted:

- **Fragmentation:** Person data is currently fragmented across administrations: every administration uses its own structures, definitions and identifiers.
- **Complicated exchange:** This fragmentation complicates data exchange between entities, because the same reality is represented and interpreted differently in each system.
- **Different representations:** Different ways of representing the same data add to the problem. The combination of KSZ/BCSS data and National Register data, for example, is often difficult in practice.

The goal of the trajectory is therefore to establish **ONE** shared, consensus-based semantic ICEG model for person data in Belgium. The following artefacts will be delivered:

Artefact	Description
Vocabulary	Formal semantic vocabulary defining classes and properties for person data
Documentation	Human-readable documentation of terms, definitions and usage guidelines
UML diagram	Visual representation of the information model and the relationships
SHACL validation rules	Validation constraints for data quality assurance
JSON-LD context file	Context file enabling JSON-based Linked Data serialisation
OpenAPI 3.0 spec	API specification for integrators/consumers to build implementation models
Code lists	Standardised reference data (nationalities, civil status codes, register types, etc.)

Questions and remarks:

Question: In which language will the model be published?

- ***Answer:*** *By default, the model is drafted and published in English only. Text values (labels and definitions) can be provided in the three national languages.*

Remark: The documented version should not only be published on GitHub, but it should also be actively sent to the French-speaking and German-speaking Communities, since this model has a clear impact on their work as well. Simply publishing on GitHub is not sufficient; it is a shared responsibility to actively promote the trajectory.

- ***Answer:*** *The link to the GitHub repository will be circulated by e-mail. The invitation list will be verified, and the participants will be asked to forward the invitation to other relevant colleagues. The trajectory will also be presented on the 'ICEG Interoperability Day' and the model will pass through the ICEG Review Committee.*

ICEG

The aim of ICEG (Interoperable Communities & eGovernment) is to model the data flows semantically and to standardise the structure of the data, in this case in the context of person data. The focus of this trajectory is explicitly on semantics: not on systems, storage or internal database implementations, but on the meaning of the data. The vocabularies and application profiles are developed in co-creation with, among others, the federal administrations (KSZ-BCSS), the Regions and the Communities, the service integrators (BOSA, MAGDA, Paradigm) and the National Register.

UML - UNIFIED MODELING LANGUAGE

Before presenting the first version of the data model, the basic concepts of UML (Unified Modeling Language) were explained. This was illustrated with the example of 'a teacher gives a course and students enrol for that course':

- **Concepts (classes):** Person, Teacher, Student, Course and Enrolment.
- **Attributes:** e.g. first name, last name, date of birth and e-mail on Person; staff number and specialisation on Teacher; student number on Student.
- **Relations (associations):** a teacher teaches 0..* courses, a course is taught by 1..* teachers, an enrolment concerns exactly 1 course. The cardinalities (1, 0..1, 0..*, 1..*) express whether a relation is mandatory or optional, single or multiple.
- **Generalisation:** Teacher and Student are subclasses of Person and inherit the person attributes, to which they add their own. A person can even be both at the same time.

OUR APPROACH

Our approach consists of several steps. We always start from use cases and we reuse elements that have already been defined. We align as much as possible with existing standards (W3C, ISO, SEMIC, OSLO, etc.) and we only define concepts ourselves where this is strictly necessary. The design of the model is iterative. After each workshop a new iteration of the model is produced: elements are added, adapted or removed on the basis of the feedback received. The report of each workshop is circulated; participants then have some time to submit comments. If no comments are received within the review period, it is assumed that what was agreed during the workshop is confirmed.

Questions and remarks:

Remark: The UML diagram presented is perceived as very strongly tied to the National Register: information type numbers (IT 000, IT 010, …) appear in the model and the reasoning follows the National Register's way of conceiving a person. The question is whether we want to put a semantic layer on top of a National Register application, or whether we want to build a broad, cross-application and cross-use-case model. Any reference to the National Register should ideally disappear from the model.

- ***Answer:*** *The information types were used as a starting point because the use cases delivered by the partners referred to them, and because it makes the model tangible and easier to grasp in a first iteration. The model itself is not tied to the National Register. It is acknowledged that the IT references should be removed from the model and its documentation. The same principle applies to the code lists: for the time being it remains useful to recognise to which information type a code list refers. For the*

scope of this trajectory, it is accepted that the notion of a person is broader than the National Register, this is precisely why a distinction is made between Person and Registered Person on the draft model. At the same time, if the model is made too broad, decisions become impossible because not all relevant stakeholders are represented. The exchange of register data therefore remains the deciding factor when assessing changes to the model, without neglecting the other aspects. The observation is not about collecting more attributes; it is about collecting the essential attributes and how these concepts are written down and named.

EXISTING STANDARDS

In order to be able to exchange information with other organisations as widely as possible, one of the best practices of a high-quality information architecture is to reuse existing data standards as much as possible. In the context of core person identification, the following standards were examined and reused:

- **SEMIC / Core Vocabularies:** The Core Person Vocabulary of SEMIC is the main basis of the model. The person class, the name structure and the identifier structure are taken over from SEMIC. The only substantial addition so far is the distinction between a Person and a Registered Person.
- **OSLO:** the existing OSLO standards were consulted, but for this trajectory it was explicitly decided to work with inter-federal URIs instead of Flanders-specific ones. For a registered person there is no European equivalent (as RegOrg exists for organisations), which is why a new class has to be introduced at the inter-federal level.

Questions and remarks:

Question: Why was a new class 'Registered Person' created and why were new attributes defined, while *dcterms:identifier* or the ADMS identifier could have been used?

- ***Answer:*** *The attribute is the ADMS identifier; only the label in the diagram is different. The label 'registration' is a translation artefact of the Dutch term 'identificator'. It was agreed that the label should be changed to 'identifier', in line with SEMIC. The class 'Registered Person' is needed because a registered person is semantically different from a person: a registered person has an identifier; a person does not necessarily have one.*

Question: Is it 'Registered Person' or 'Belgian Registered Person'? Has portability been considered, for people moving across borders, in Europe and beyond?

- ***Answer:*** *Semantically the model allows a foreign identifier to be used. In reality, a person moving to Belgium currently receives a Belgian number, but that is a process and legal matter and not the scope of this semantic model. The model must be open enough to accommodate a change in legislation. The model should allow multiple Identifiers.*

DRAFT MODEL

On the basis of the use cases collected during the business workshop, the draft model is built up step by step. The scope of the trajectory as a whole is the following:

In scope	Out of scope
Natural Person: identity, identifiers (SSIN), name, date/place of birth, gender, nationality, domicile, residence, contact information, civil status, legal capacity	Domain-specific person profiles (health, taxation, profession, etc.)
Registered Person: registration in official registers (National Register, BIS register)	Authentication and authorisation mechanisms
Links to other domains: address (aligned with BeSt Address), possible relation to organisation (via CBE), jurisdiction, household composition, filiation	Internal database properties, schemas or storage implementations

The scope of this first thematic workshop covered the following information types:

- Identification number (IT 000)
- Reference file (IT 002)
- Name, first names, titles, aliases and pseudonyms (IT 010, 011, 012, 013)
- Birth information (IT 100, 101)
- Gender / registered sex (IT 004)

Questions and remarks:

Question: Is historical information in scope? A person may have a history of names; the data evolves over time from birth to death, and so do the relations between persons.

- ***Answer:*** *During the kick-off it was agreed that purely technical database changes (insert/update/delete operations in a back-end system) are out of scope, whereas real-life events (a change of address, a change of name, a change of registered sex) are in scope. It is acknowledged that the boundary between a 'real-life event' and a 'database event' is difficult to draw in practice, a correction of a typing error, for instance, is not a real-life event but it does have to be exchanged. Changes in reality are what belongs in an exchange standard. The model should also be open enough to absorb a change of legislation within a few years.*

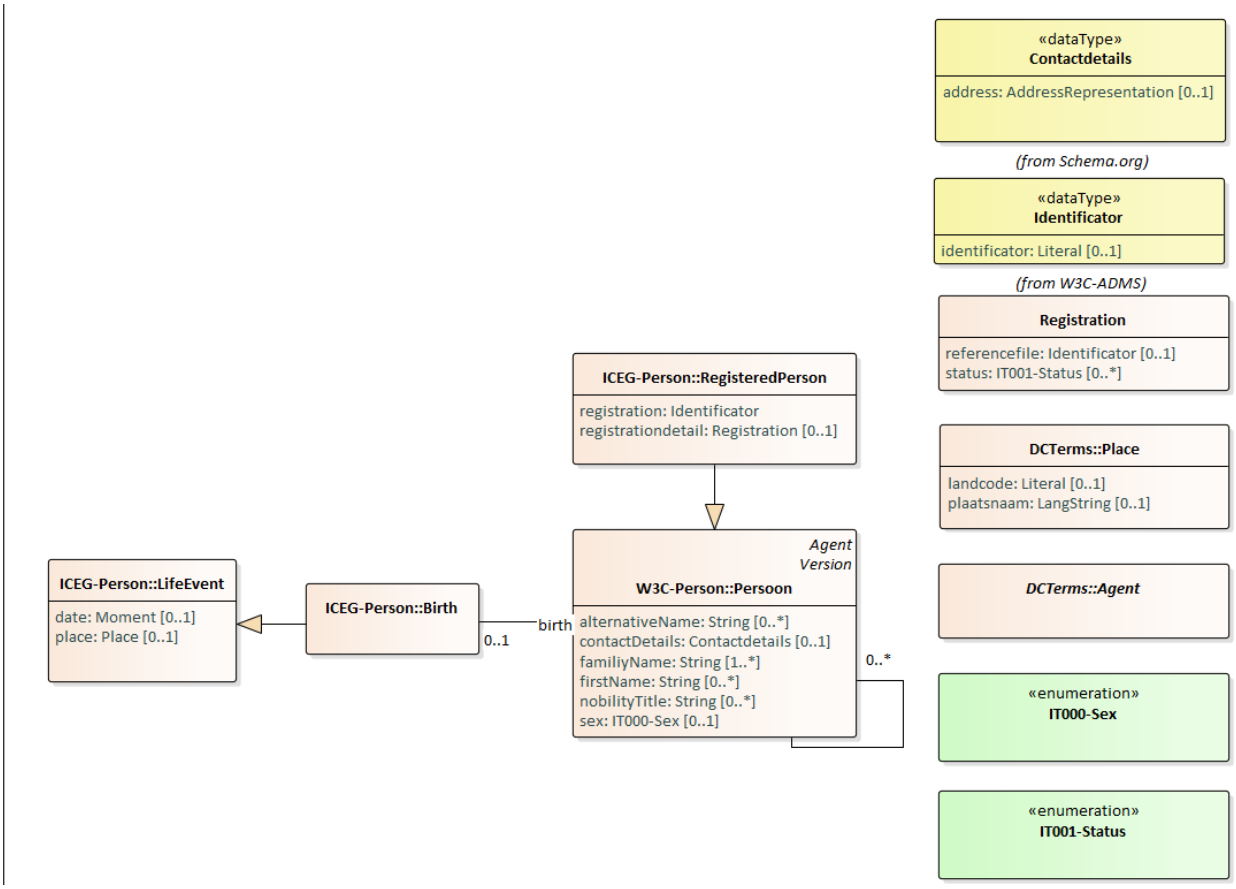
Question: Are fictitious persons in scope?

- ***Answer:*** *Fictitious persons are out of scope for this trajectory.*

Question: Why is Nationality not captured in draft model?

- ***Answer:*** *Nationality is formally scheduled for the fourth thematic workshop, but as it returns in nearly every topic it should have been added to this workshop.*

Below is an overview of the draft model presented during the First Thematic Workshop.



DISCUSSION

The draft model was built up based on three use cases covering the agreed scope for this thematic workshop. The main classes of the draft model and the feedback/questions raised during this first thematic working group tackling the Core Identification of a person are discussed in more detail below.

USE CASE 1 - Registration of a person

Julie is born in the hospital of Leuven. Her parents declare the birth to the local authority. The municipality registers her name, her first names and her birth data, and an identification number is assigned to her.
The first use case concerns the very first registration of a natural person. Julie is born; her parents declare the birth to the local authority. A Declaration is the formal act by which the parents notify the local authority of a birth, and it triggers the registration of the Name and the Birth information.

The following classes and information types were presented:

- **Person:**
  - **FirstName:** containing a person's first name.
  - **LastName**: containing a person's family name. In the draft model, familyName and firstName are modelled as string attributes on Person, taken over from SEMIC.
  - **Sex**: the registered sex, currently supporting only 'female' and 'male' in the National Register. In the draft model this is modelled as an enumeration / code list, so that it can be extended.

- **Life event: isBirth**
  - **place:** For births in Belgium the NIS code of the municipality is used; for births abroad, the place name plus the country code.
  - **date:** the information type holding the date of birth and the type of supporting document used to prove it.
  - **Supporting documents**

In our use case, based on the registered Name and Birth data, a unique Identification Number is assigned. The identification number is the unique 11-digit number (SSIN/NI) that is automatically assigned upon a person's first registration, computed from the date of birth, the sex and a control number. Once cancelled, an identification number can never be reused. In the draft model a distinction is made between Person and Registered Person. A Registered Person is a Person who carries an identifier.

Questions/remarks:

Remark: There is a typing error in the model: 'familiyName' must read 'familyName'.

- ***Answer:*** *Agreed; this will be corrected in the next iteration.*

Remark: The model must explicitly allow more than one identifier per person. Identifiers are context-specific: next to the SSIN and the BIS number, there are for example the student/pupil identifiers used by the Communities, sector-specific numbers, foreign identifiers (e.g. a Danish national number), passport numbers and Web IDs (used in Solid-based projects, where a permanent Web ID has to travel with the data and has to be mapped when a national identifier changes). A semantic model should state that one person can be identified by many identifiers.

- ***Answer:*** *The cardinality between (Registered) Person and Identifier will be changed so that multiple identifiers are possible, including foreign identifiers that can be linked to the same person.*

Remark: If multiple identifiers are allowed, metadata on the identifier is indispensable. Without it, five numbers can be present without knowing who issued them. The following metadata must be added: issuer (which authority assigned the number, commercial or governmental), type/scheme of the identifier, and the period of validity (start date and end date).

- ***Answer:*** *Only the minimum required by the use case was shown in this first draft; issuer, type and validity will be added by reusing the SEMIC/ADMS identifier structure.*

Remark: Who registers first? Today the declaration to the civil status officer leads to the registration in the National Register, while the first registration in practice often happens in the hospital via the Crossroads Bank for Social Security. Projects are ongoing to register directly in the National Register from the hospital.

- **Answer:** For the semantic model it does not matter through which channel the registration arrives.

Remark: A person should ideally keep one and the same number from birth to death. The only exception is the situation in which several files exist for one person and they must be merged.

USE CASE 2 - Registration of a noble title and a pseudonym

Philippe is granted the noble title of "Knight" by Royal Decree. The municipality registers his new title, referencing the decree. He also uses a pseudonym in his artistic career.

Noble title

A Noble Title is an honorary distinction assigned to a person (e.g. Knight, Baron) together with the date on which the title was obtained. In the draft model the noble title is currently modelled in the same way as the alternative name, which is why both topics were discussed together.

Pseudonym, alias and declared name

A Pseudonym is an assumed name, independent of the legal name, most often adopted by artists. It has no legal value for official identification, but it can still be protected in the same way as a name. In the current draft model this is captured by a single attribute 'alternativeName' on Person, modelled as a string.

Questions/remarks:

Remark: Is 'alternativeName' the attribute that will be used for pseudonyms? If so, a pseudonym can itself consist of a family name and first names. It would therefore be better to introduce a name type (a separate class or complex data type) with family name and first names, which can be used both for the normal name of the person and for the pseudonym. Otherwise it has to be documented explicitly that all components of the pseudonym are concatenated into one 'alternativeName' string.

- ***Answer:*** *Confirmed as a point of attention. Both the official name and the alternative name are currently modelled as simple string attributes, which might not be sufficient for all use cases. Names are more complicated than a string (double family names combining the mother's and the father's name, patronymic and matronymic name forms, middle names, etc.).*

Question: What exactly is the difference between a pseudonym and an alias? Both notions exist and they are not interchangeable: a pseudonym is for example the stage name of an artist, whereas an alias may be a full substitute identity (for example a protected person who had to relocate). In the current registers this is spread over different information types, which makes the relation between them unclear.

- ***Answer:*** *The relation between the two information types must be examined. The proposal is not to copy the register structure but to take pseudonym, alias and declared name together under one broader concept of 'other names', and to distinguish the individual cases by means of a type and additional attributes.*

Remark: A fundamental distinction that is currently missing in the registers, and which should be raised to the level of the model, is the distinction between information based on a legal document and information that is merely declared. For a foreigner who arrives without official documents, the name is registered as 'declared'. The proposal is to add this as an attribute as it must be exchanged. A civil status officer who draws up a document needs to know whether the identity is legally founded or only declared. A 'declared' status can evolve into a legally founded one: a name that is declared today may be supported by a passport two months later, and the source then changes. This again demonstrates the need for historic tracking.

Remark: The artist's name can be considered domain-specific and out of scope for this trajectory. Artist names are registered and controlled in separate (authors') registers. The suggestion was made to drop the artist name from the model, or alternatively to treat it as just another identifier (an identifier for the artistic domain) outside of name and first name. Counter-arguments were also raised: some artists are known by their artist name throughout their life and only use their given name on a handful of official documents; in the context of a possible obligation for content creators to register in the Crossroads Bank for Enterprises, the artist name may become a relevant use case; and the same mechanism also covers calling names that are not artist names at all (e.g. a person officially registered as Maria but addressed by a completely different name, or a person who uses a first name other than their registered first name - which is common in several cultures).

Conclusion: The generic concept 'alternative name' (or 'other names') is kept because it is broader than the register-specific notions, but it must not remain a plain string. Attributes should be added so that the same construct can be used in different situations:

- family name and first names (structured, instead of one concatenated string);
- type of other name (pseudonym, alias, calling name, artist name);
- source / origin of the information / issuer (e.g. a foreign register, the artistic domain, a declared);
- legal basis (legal document versus declared);
- period of validity (start date and end date).

The same reasoning applies to the noble title: a noble title can be acquired during a person's life and can also be lost, so it must carry its own metadata (at least the date of acquisition and a validity period) and it must be possible to keep the earlier titles.

USE CASE 3 - Change of name and registered sex

Jonas was registered at birth under a different first name and a different sex. After a procedure with the competent authority, his first name is changed and his registered sex is adapted. The previous situation must remain traceable.

Three information types are relevant for this storyline:

- **NameModification:** the information type recording the reason for a change or a correction of the name or first names. It is always dated and justified by means of a reason code. It explicitly covers modifications of the name, the first names and the noble title with the same structure (date + reason code + comment).
- **GenderModification:** the information type recording the modification of a person's registered sex.

- **ReferenceFile:** the information type recorded in a cancelled file, linking it to the new file that replaces it. It contains the date of annulment, the identification number of the new registered person and a code explaining the reason for the cancellation; the reference file ensures traceability between the old and the new file. In the current draft model, which follows the way the National Register works today, a change of first name and registered sex results in the creation of a new registered person, while the old file is cancelled and points to the new one through the reference file. This approach is presented as a starting point and is open to discussion.

Questions/remarks:

Remark: Gender and sex must be modelled separately. Semantically, the registered sex and the gender are two different things. Gender is mainly used for salutations, which is included under 'contact details' as well. It was also noted that in some domains (e.g. culture) the attribute is no longer included at all because it is no longer meaningful, and that within the National Register modernisation it is being questioned whether sex/gender should still be registered for identification purposes. Whether a separate 'sex at birth' should be retained next to the registered sex and the gender was discussed. This was considered relevant from a health perspective, and it was argued that this is domain-specific and out-of-scope. Additionally, the sex registered at birth can be derived from historic tracking.

Remark: The distinction between a correction and a real-life change is missing in the model. A correction (e.g. a name spelled with an 'O' instead of an 'R', or a missing letter) is not a real-life change of name: the corrected value applies from the same start date as the original registration, because the original value was simply wrong. A real-life change of name, on the other hand, starts on a new date and ends the previous situation. The model should make it possible to distinguish these two situations without the consumer having to interpret dates, and to know from which date a value must be acted upon. It remains valuable to keep the value before correction, because older documents will still show the old value.

- ***Answer:*** *The reason for the change must therefore be recorded together with the change itself. It was also noted that in the register the notions of 'end date' and 'deletion date' are currently very difficult to distinguish, which makes the data difficult to interpret: a deletion should mean that the information never existed, while an end date means that the information was valid until a certain moment. This distinction should be visible in the model.*

Remark: Historic tracking is missing from the current draft model. Time is an important element when speaking about person data: every element (name, sex, noble title, …) has its own life cycle and can evolve independently. Concrete needs mentioned: retrieving the name a person had at a given point in time in order to interpret an old legal document; contacting the person responsible for a bankrupt enterprise whose name has changed in the meantime; linking old documents to the current person.

- ***Option 1****: Add validity metadata (start date, end date, reason, source, legal basis) to the individual attributes or to the separate classes that hold them.*
- ***Option 2****: Every time an attribute changes, create a new version of the person and link the versions together, so that a consumer can ask for the version of the person as it was at a given point in time.*
- ***Answer****: Each approach has its own merits and downsides. Versioning is editorially easier than putting all the validity metadata in the model, and it makes it straightforward to retrieve 'the person as of date X'. Adding validity metadata to every attribute, can create a lot of noise in the model and risks obscuring the essence of the model. Versioning driven by 'any attribute change' can become a*

challenge as a model can have many changing attributes and many relations creating a large set of possible version increments.

Remark: The reference file mechanism should not be used as the basis for a change of first name and registered sex. Using a reference file means creating a new registered person, which is semantically not correct and complicates any historical tracking.

- ***Answer:*** *A link between two identification numbers will nevertheless always be needed, for historical reasons or for example when a person who first received a BIS number later obtains a national number, or when it is discovered that two files in fact concern the same person and they must be merged. The core team will prepare an alternative proposal for expressing the traceability link between a 'cancelled' and a 'replacing' registration of a person as an alternative to the reference file modelled as it exists today.*

Remark: If a link between two files is modelled, in which direction should it point: from the old to the new, or from the new to the old? Both were reported to occur in practice. Normally the old identifier points to the newer one, so that a lookup on the old identifier leads to the current person. In some situations, however, the old number is retained because more institutions know it, while in other situations (e.g. for FAS/eID) the new number must be used. This has a real impact on the consuming institutions

- ***Answer:*** *From the point of view of the model, the direction must be made explicit, and a status must be used to indicate which identifier is active and which is cancelled. The core team will include this in the proposal, with the ambition to make the model simpler than current practice, while remaining able to represent what can happen.*

Remark: The cardinality between Identification and Reference File (0..1) is too restrictive: a person can have several cancelled files and several national numbers, and these can be chained (A to B, B to C, and even reversed afterwards). Changes can occur more than once, for the registered sex as well as for the date of birth.

- ***Answer:*** *The cardinalities will be adapted and a self-association on the registration / registered person will be used to express the chain between the successive registrations.*

4. NEXT STEPS

All input from the thematic working group has been consolidated in this report, for which any feedback is welcome. The input and feedback gathered during the workshop and review period will be incorporated into the next iteration of the draft model. Feedback is captured via GitHub.

The second thematic workshop will take place on Monday 19 October 2026, from 10:00 to 12:30, at the same location. The topic is ***Address and Residence***. Registration is possible via this link. Participants are invited to forward the invitation to other colleagues who are relevant for the topics to be addressed.

The date of the fourth thematic workshop was moved from 10 December to ***Monday 14 December 2026 (10:00-12:30)***, because of a conflicting workshop on the morning of 10 December for another trajectory.